

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2019-2020)

PRE-MEDICAL : ENTHUSIAST COURSE [PHASE-I]

TEST # 18

DATE : 09-01-2020

REVISION TEST

Test Pattern : NEET

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **720**.
5. There are **three** parts in the question paper A,B,C consisting of **Physics, Chemistry** and **Biology** having **45 questions** each in Physics & Chemistry and **90 questions** in Biology part of equal weightage. Each question is allotted 4 (four) marks for **correct** response.
6. **One Fourth** mark will be deducted for indicated incorrect response of each question. **No deduction** from the total score will be made if no response is indicated for an item in the Answer Sheet.
7. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
8. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
9. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
10. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
11. **Do not fold or make any stray marks on the Answer Sheet.**

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REVISION TEST # 18

Topics

DATE : 09 - 01 - 2020

PRE-MEDICAL ENTHUSIAST COURSE : PHASE - I – PATTERN : [NEET]

PHYSICS : Semiconductors

CHEMISTRY : Biomolecules, Chemistry in Every Day Life + IUPAC

BOTANY : Transport, Mineral Nutrition < Plant Growth & Development, Plant Reproduction

SPACE FOR ROUGH WORK

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

- | | |
|---|---|
| <p>1. The de-broglie wavelength of electron in the first Bohr orbit is :</p> <p>(1) Equal to one-fourth of circumference of first orbit</p> <p>(2) Equal to half of circumference of first orbit</p> <p>(3) Equal to twice of circumference of first orbit</p> <p>(4) Equal to circumference of first orbit</p> <p>2. In Bohr model of hydrogen, the force on electron depends on the principal quantum number as :</p> <p>(1) $F \propto \frac{1}{n^2}$</p> <p>(2) $F \propto \frac{1}{n^4}$</p> <p>(3) $F \propto \frac{1}{n^5}$</p> <p>(4) $F \propto \frac{1}{n}$</p> <p>3. If intensity of incident light in photoelectric emission is increased keeping the frequency constant, the which of the following is TRUE?</p> <p>(1) Maximum kinetic energy of ejected electrons will increase</p> <p>(2) Work function remains unchanged</p> <p>(3) Stopping potential will decrease</p> <p>(4) Maximum kinetic energy of ejected electron will decrease</p> | <p>4. When a metal surface is illuminated by light of wavelength 400 nm and 250 nm, the maximum speed of photoelectrons are v and $2v$ respectively. The work function of metal is :</p> <p>(1) $2hc \times 10^6 \text{ J}$</p> <p>(2) $1.5 hc \times 10^6 \text{ J}$</p> <p>(3) $hc \times 10^6 \text{ J}$</p> <p>(4) $0.5 hc \times 10^6 \text{ J}$</p> <p>5. In the photoelectric experiment, fastest electron has speed v if wavelength used is λ. If wavelength used is $\frac{3\lambda}{4}$, the speed of fastest electron will be :-</p> <p>(1) $\left(\frac{3}{4}\right)^{\frac{1}{2}} v$</p> <p>(2) $\left(\frac{4}{3}\right)^{\frac{1}{2}} v$</p> <p>(3) Less than $\left(\frac{4}{3}\right)^{\frac{1}{2}} v$</p> <p>(4) Greater than $\left(\frac{4}{3}\right)^{\frac{1}{2}} v$</p> |
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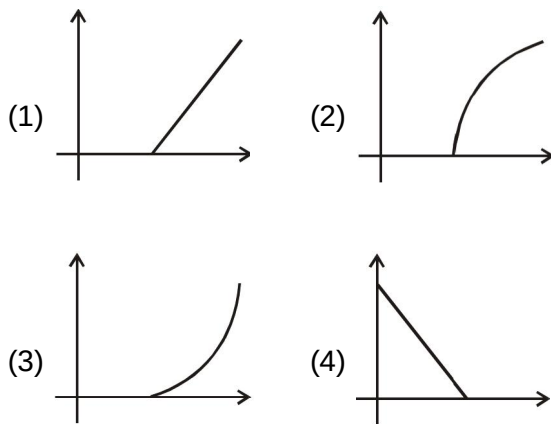
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6. Energy level A, B and C of a certain atom corresponding to increasing values of energy i.e. $E_A < E_B < E_C$. If λ_1, λ_2 and λ_3 are wavelength of radiations corresponding to transition $C \rightarrow B$, $B \rightarrow A$ and $C \rightarrow A$ respectively, then choose CORRECT relation :-

(1) $\lambda_3 = \lambda_1 + \lambda_2$ (2) $\lambda_3 = \frac{\lambda_1 \lambda_2}{\lambda_1 + \lambda_2}$

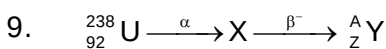
(3) $\lambda_1 + \lambda_2 + \lambda_3 = 0$ (4) $\lambda_3^2 = \lambda_1^2 + \lambda_2^2$

7. The graph representing correct relation of frequency of characteristic X-ray with the atomic number Z of material is :



8. The wavelength of k_α line for an element of atomic no. 43 is λ . The wavelength of k_α line for an element of atomic no 29 is :-

(1) $\frac{43}{29}\lambda$ (2) $\frac{42}{28}\lambda$ (3) $\frac{9}{4}\lambda$ (4) $\frac{4}{9}\lambda$



The value of Z and A respectively will be :

(1) 92, 236 (2) 88, 230
(3) 90, 234 (4) 91, 234

10. During mean life of radioactive element, the fraction that disintegrates is :-

(1) e (2) $\frac{1}{e}$ (3) $\frac{e-1}{e}$ (4) $\frac{e}{e-1}$

11. A radioactive element has half-life of 5 years. The fraction of atoms that would decay in 15 year will be :-

(1) $\frac{1}{8}$ (2) $\frac{2}{3}$ (3) $\frac{7}{8}$ (4) $\frac{5}{8}$

12. Energy released in fission of single ${}_{92}\text{U}^{235}$ nucleus is 200 MeV. The fission rate of ${}_{92}\text{U}^{235}$ fuelled reactor operating at power of 5W is :-

(1) $1.56 \times 10^{10} \text{ s}^{-1}$
(2) $1.56 \times 10^{11} \text{ s}^{-1}$
(3) $1.56 \times 10^{16} \text{ s}^{-1}$
(4) $1.56 \times 10^{17} \text{ s}^{-1}$

13. The binding energy of α -particle ${}^4_2\text{He}$ is 7.047 MeV per nucleon and the binding energy of deuteron ${}^2_1\text{H}$ is 1.112 MeV per nucleon. Then in the fusion reaction ${}^2_1\text{H} + {}^2_1\text{H} \rightarrow {}^4_2\text{He} + Q$, the energy Q released is

(1) 23.74 MeV (2) 32.82 MeV
(3) 11.9 MeV (4) 4.94 MeV

14. One requires energy E_n to remove a nucleon from a nucleus and an energy E_e to remove an electron from the orbit of an atom. Then

(1) $E_n > E_e$ (2) $E_n < E_e$
(3) $E_n = E_e$ (4) $E_n \geq d E_e$

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15. Radius of ${}^4_2\text{He}$ nucleus is 3 Fermi. The radius of ${}^{32}_{16}\text{S}$ nucleus will be nearest to
 (1) 6 Fermi (2) 4 Fermi
 (3) 5 Fermi (4) 8 Fermi
16. Two nucleons are at a separation of 1×10^{-15} m. The net force between them is F_1 if both are protons, F_2 if both are neutrons and F_3 if one is a proton and other is a neutron. In such a case
 (1) $F_3 = F_2 > F_1$ (2) $F_1 = F_2 > F_3$
 (3) $F_1 = F_3 > F_2$ (4) $F_2 > F_1 > F_3$
17. If m , m_n and m_p are the masses of ${}_Z\text{X}^A$ nucleus, neutron and proton respectively
 (1) $m = (A - Z) m_n + Z m_p$
 (2) $m = (A - Z) m_p + Z m_n$
 (3) $m < (A - Z) m_n + Z m_p$
 (4) $m > (A - Z) m_n + Z m_p$
18. The control rod in a nuclear reactor is made of
 (1) Graphite (2) Plutonium
 (3) Cadmium (4) Uranium
19. Nucleus A divides into two nuclei B and C in a fission process, their binding energies being E_a , E_b and E_c respectively. Then
 (1) $E_b + E_c > E_a$ (2) $E_b + E_c < E_a$
 (3) $E_b + E_c = E_a$ (4) $E_b - E_c = E_a$
20. A reactor is generating 1000 kW of power and 200 MeV of energy may be obtained per fission of U^{235} . The rate of nuclear fission in the reactor is
 (1) 3.125×10^{16} (2) 1.253×10^{16}
 (3) 2×10^8 (4) 931
21. When ${}_{92}\text{U}^{235}$ undergoes fission by absorbing ${}_0^1\text{n}$ and ${}_{56}\text{Ba}^{144}$ and ${}_{36}\text{Kr}^{89}$ are formed. The number of neutrons produced will be
 (1) 2 (2) 3 (3) 1 (4) 0
22. A nucleus of ${}_{84}\text{Po}^{210}$ originally at rest emit α -particle with speed v . Recoil speed of the daughter nucleus is
 (1) $\frac{v}{210}$ (2) $\frac{v}{84}$ (3) $\frac{4v}{206}$ (4) $\frac{3v}{214}$
23. Which of the following is most unstable
 (1) α -particle
 (2) Neutrons
 (3) β^- -particle
 (4) β^+ -particle
24. In a radioactive substance, initially 8×10^4 number of atoms are there, having half life period 3 years. The number of atoms 1×10^4 will remain after interval of
 (1) 6 years (2) 24 years
 (3) 3 years (4) 9 years
25. An α -particle colliding with one of the electrons in a gold atom loses
 (1) Most of its momentum
 (2) About $\frac{1}{3}$ rd of its momentum
 (3) Little of its energy
 (4) Most of its energy
26. The angular momentum of an electron in a hydrogen atom is proportional to (where n is principle quantum number)
 (1) n (2) n^2
 (3) n^3 (4) $\sqrt{n}$

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| <p>27. When an electron in hydrogen atom is taken from fourth excited state to ground state</p> <p>(1) Both kinetic energy and potential energy increases
 (2) Both kinetic energy and potential energy decreases
 (3) Kinetic energy will increase while potential energy will decrease
 (4) Kinetic energy will decrease while potential energy will increase</p> <p>28. What should be the angular momentum of an electron in Bohr's hydrogen atom whose energy is -0.544 eV?</p> <p>(1) $\frac{h}{\pi}$ (2) $\frac{3h}{2\pi}$ (3) $\frac{5h}{2\pi}$ (4) $\frac{2h}{2\pi}$</p> <p>29. Ground state energy of H-atom is -13.6 eV. The energy needed to ionise H-atom from its second excited state is</p> <p>(1) 1.51 eV (2) 3.4 eV
 (3) 13.6 eV (4) 12.1 eV</p> <p>30. The product of angular speed and tangential speed of electron in n^{th} orbit of hydrogen atom is</p> <p>(1) Directly proportional to n^2
 (2) Directly proportional to n^3
 (3) Inversely proportional to n^4
 (4) Independent of n</p> <p>31. What should be the ratio of minimum to maximum wavelength of radiation emitted by transition of an electron to ground state of Bohr's hydrogen atom?</p> <p>(1) $\frac{3}{4}$ (2) $\frac{1}{4}$ (3) $\frac{1}{8}$ (4) $\frac{3}{8}$</p> | <p>32. The ratio of energies of hydrogen atom in its first excited state to third excited state is</p> <p>(1) $\frac{1}{4}$ (2) $\frac{4}{1}$ (3) $\frac{3}{4}$ (4) $\frac{4}{3}$</p> <p>33. From coolidge tube experiment, X-rays are used to detect fracture present in bones. The wavelength range required to detect fracture is $1.24 \text{ \AA}$ to $2.48 \text{ \AA}$. Find minimum accelerating voltage required to operate given X-ray machine :
 (Given : $hC = 12400 \text{ eV\AA}$)</p> <p>(1) 10000 Volt (2) 5000 Volt
 (3) 10520 Volt (4) 7440 Volt</p> <p>34. If the wavelength of first member of Lyman series is λ then calculate the wavelength of first member of Pfund series</p> <p>(1) $\frac{675}{11}\lambda$ (2) $\frac{245}{11}\lambda$
 (3) $\frac{322}{13}\lambda$ (4) $\frac{289}{11}\lambda$</p> <p>35. In Bohr's model of the hydrogen atom, the ratio between the period of revolution of an electron in the orbit of $n = 1$ to the period of revolution of the electron in the orbit $n = 2$ is</p> <p>(1) $2 : 1$ (2) $1 : 2$ (3) $1 : 4$ (4) $1 : 8$</p> <p>36. If an electron in hydrogen atom jumps from third orbit to second orbit, the frequency of the emitted radiation is given by (c is speed of light)</p> <p>(1) $\frac{3Rc}{29}$ (2) $\frac{5Rc}{36}$ (3) $\frac{7Rc}{36}$ (4) $\frac{8Rc}{31}$</p> |
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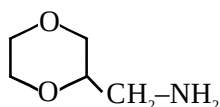
37. Let F_1 be the frequency of second line of Lyman series and F_2 be the frequency of first line of Balmer series then frequency of first line of Lyman series is given by
 (1) $F_1 - F_2$ (2) $F_1 + F_2$
 (3) $F_2 - F_1$ (4) $\frac{F_1 F_2}{F_1 + F_2}$
38. If the difference between $(n + 1)^{\text{th}}$ Bohr radius and n^{th} Bohr radius is equal to the $(n - 1)^{\text{th}}$ Bohr radius then find the value of n
 (1) 4 (2) 3 (3) 2 (4) 1
39. The lines in Balmer series have their wavelengths lying between
 (1) 1266 Å to 3647 Å
 (2) 642 Å to 3000 Å
 (3) 3647 Å to 6563 Å
 (4) Zero to infinity
40. When radiation is incident on a photoelectron emitter, the stopping potential is found to be 9 volts. If e/m for the electron is $1.8 \times 10^{11} \text{ C kg}^{-1}$ the maximum velocity of the ejected electrons is :-
 (1) $6 \times 10^5 \text{ ms}^{-1}$
 (2) $8 \times 10^5 \text{ ms}^{-1}$
 (3) $1.8 \times 10^6 \text{ ms}^{-1}$
 (4) $1.8 \times 10^5 \text{ ms}^{-1}$
41. If longest wavelength of Balmer series of H atom is λ then shortest wavelength of Lyman series will be
 (1) $\frac{5}{36}\lambda$ (2) $\frac{5}{9}\lambda$ (3) $\frac{36}{5}\lambda$ (4) $\frac{9}{5}\lambda$
42. An electron in a hydrogen atom makes a transition such that its kinetic energy increases, then
 (1) The electron may have excited form $n = 2$ to $n = 3$
 (2) Potential energy of the electron increases
 (3) Potential energy of the electron decreases
 (4) Total energy of the electron increases
43. The magnetic field at the centre of hydrogen atom due to the motion of the electron in the first Bohr orbit is B . The magnetic field at the centre due to motion of the electron in 2^{nd} orbit is
 (1) $\frac{B}{4}$ (2) $\frac{B}{8}$
 (3) $\frac{B}{32}$ (4) $\frac{B}{64}$
44. A hydrogen atom in ground state absorbs 12.09 eV of energy. The orbital angular momentum of the electron
 (1) Is doubles
 (2) Is halved
 (3) Remains same
 (4) Becomes three times
45. When a point source of light is at a distance of one metre from a photo cell, the cut off voltage is found to be V . If the same source is placed at 2 m distance from photo cell, the cut off voltage will be :-
 (1) V (2) $V/2$
 (3) $V/4$ (4) $V/\sqrt{2}$

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46. Which of the following statements is/are true about a homologous series :-

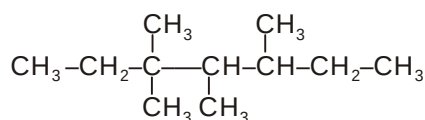
- (1) Different homologous series may have same general formula
- (2) Alkane homologous series have C_nH_{2n} general formula
- (3) Homologues have same physical but different chemical properties
- (4) All of them

47. Number of hetero atoms present in the given compound is :



- (1) 2 (2) 3 (3) 4 (4) 1

48. In following compound



The correct lowest set of locant is :

- (1) 3,3,4,5 (2) 3,4,5,5
(3) 4,5,3,3 (4) 5,5,4,3

49. Which of the following compound have maximum number of olefinic bond :-

- (1) $H_2C=CH-CH=O$
- (2) $HO-C(=O)-CH=CH-C(=O)-H$
- (3) $H_2C=CH-C\equiv C-CH=CH_2$
- (4) $H_3C-C\equiv C-CH_3$

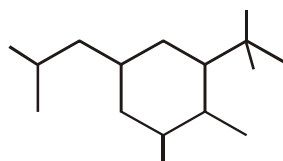
50. Tertiary butyl ethyl amine is -

- (1) 1° Amine (2) 2° Amine
(3) 3° Amine (4) Quaternary salt

51. Vinyl carbinol is -

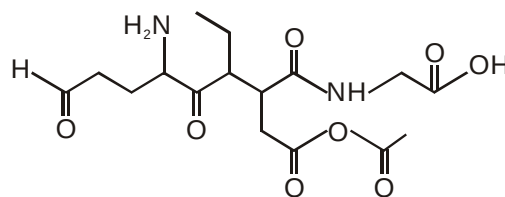
- (1) $HO-CH_2-CH=CH_2$
- (2) $CH_3-C(OH)=CH_2$
- (3) $H_3C-CH=CH-OH$
- (4) $H_3C-C(CH_2-OH)=CH_2$

52. Number of 1°, 2°, 3° and 4° C present in compound are respectively



- (1) 7, 4, 5 and 1 (2) 7, 3, 4 and 1
(3) 7, 3, 5 and 1 (4) 7, 4, 4 and 1

53. The number of functional groups present in the following compounds is :



- (1) 5 (2) 7 (3) 6 (4) 8

54. The structure of isobutyl group in an organic compound is :-

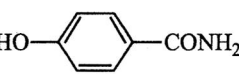
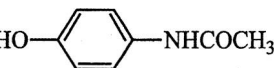
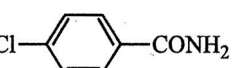
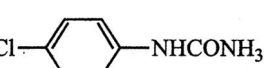
- (1) $CH_3-C(CH_3)_2-$
- (2) $CH_3-CH(CH_3)-CH_2-$
- (3) $CH_3-CH_2-CH_2-$
- (4) $CH_3CH_2CH_2CH_2-$

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| <p>55. Sucrose on hydrolysis gives:
 (1) glucose only
 (2) glucose and galactose
 (3) glucose and fructose
 (4) glucose and lactose</p> <p>56. The disaccharide present in milk is:
 (1) sucrose (2) lactose
 (3) maltose (4) none of these</p> <p>57. The change in optical rotation with time of freshly prepared solutions of sugar, is known as:
 (1) specific rotation (2) inversion
 (3) rotatory motion (4) mutarotation</p> <p>58. Helical structure of proteins is stabilized by:
 (1) ionic bond
 (2) covalent bond
 (3) van der Waals' forces
 (4) hydrogen bonds</p> <p>59. Vitamin C is:
 (1) ascorbic acid (2) lactic acid
 (3) citric acid (4) paracetamol</p> <p>60. When glucose is reacted with bromine water, the major product is:
 (1) gluconic acid (2) saccharic acid
 (3) sorbitol (4) galactose</p> <p>61. Glucose and fructose are:
 (1) optical isomers
 (2) functional group isomers
 (3) chain isomers
 (4) position isomers</p> <p>62. Which polymers occur naturally?
 (1) Starch and Nylon
 (2) Starch and Polythene
 (3) Proteins and Nylon
 (4) Proteins and Natural Rubber</p> | <p>63. Which of the following have branched structure?
 (1) Low density polythene
 (2) High density polythene
 (3) Amylose
 (4) Cellulose</p> <p>64. Which statement about aspirin is not true
 (1) Aspirin belongs to narcotic analgesics.
 (2) It is effective in relieving pain.
 (3) It has anti blood clotting action.
 (4) It is a neurologically active drug.</p> <p>65. Thermosetting polymers are
 (1) cross-linked polymers
 (2) do not melt or soften on heating
 (3) cross-linking occurs during heating when it hardens irreversibly
 (4) all are correct.</p> <p>66. Which one of the following is used to make 'nonstick' cookware ?
 (1) PVC
 (2) Polystyrene
 (3) Poly (ethylene terephthalate)
 (4) Polytetrafluoroethylene</p> <p>67. Which compound/set of compounds is used in the, manufacture of Nylon-6, 6 ?
 (1) $\text{HOOC}(\text{CH}_2)_4\text{COOH} + \text{H}_2\text{N}(\text{CH}_2)_6\text{NH}_2$
 (2) $\text{CH}_3 = \text{CH}-\text{C}(\text{CH}_3) = \text{CH}_2$
 (3) $\text{CH}_2 = \text{CH}_2$
 (4) $\text{HOOC}-\text{C}_6\text{H}_4-\text{COOH} + \text{HOCH}_2-\text{CH}_2\text{OH}$</p> <p>68. What percentage of sulphur is used in the vulcanization of rubber?
 (1) 5% (2) 3% (3) 30% (4) 55%</p> |
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69. The correct structure of the drug paracetamol is

- (1) 
- (2) 
- (3) 
- (4) 

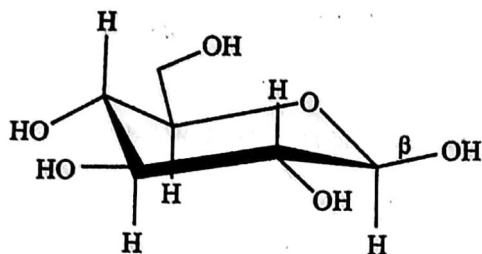
70. The compound that causes general antidepressant action on the central nervous system belongs to the class of

- (1) analgesics
(2) tranquilizers
(3) narcotic analgesics
(4) antihistamines

71. Which is a reducing sugar

- (1) Glucose
(2) Fructose
(3) Galactose
(4) All of these

72. The following carbohydrate is

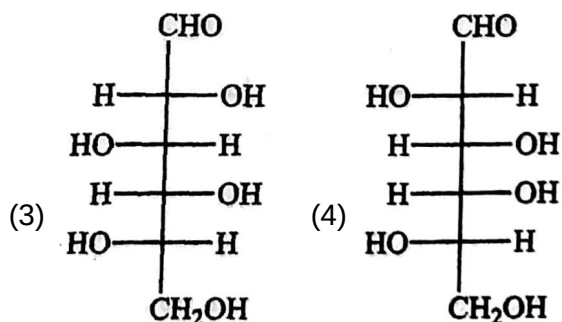
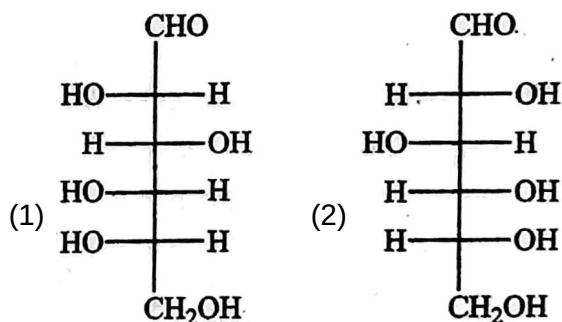


- (1) a ketohexose
(2) an aldohexose
(3) an α -furanose
(4) an α -pyranose

73. Glycerol is added to soap. It functions

- (1) as a filler.
(2) to increase leathering.
(3) to prevent rapid drying.
(4) to make soap granules.

74. Which of the following Fischer projections represents L-glucose?



75. The number of chiral carbons in β -D (+) glucose is

- (1) five
(2) six
(3) three
(4) four

76. Thermoplastics are

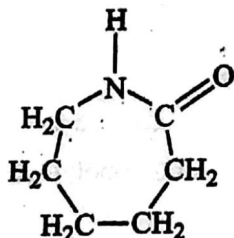
- (1) linear polymers
(2) soften or melt on heating
(3) molten polymer can be moulded in desired shape
(4) all the correct.

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77. Which of the following is a step-growth polymer?

- (1) Styrene (2) Nylon-6, 6
(3) Teflon (4) Rubber

78. Which of the following polymer can be formed by using the following monomer unit?



- (1) Nylon 6, 6 (2) Nylon 2-nylon 6
(3) Melmac (4) Nylon-6

79. Which biomolecule is distributed more widely in a cell?

- (1) Chloroplast (2) RNA
(3) DNA (4) Spaherosomes

80. Most abundant RNA in the cell

- (1) r-RNA
(2) m-RNA
(3) t-RNA
(4) t-RNA threonine

81. Name the simplest amino acid

- (1) Alanine (2) Tyrosine
(3) Asparagine (4) Glycine

82. Mineral associated with cytochrome is

- (1) Mg (2) Cu and Ag
(3) Fe (4) Cu

83. The term enzyme was coined by

- (1) Urey Miller (2) Pasteur
(3) Kuhne (4) Buchner

84. β -oxidation occurs in

- (1) Nucleus
(2) Cytoplasm
(3) Mitochondria
(4) Chloroplast

85. High content of triglycerides are found in

- (1) VLDL (2) LDL
(3) HDL (4) Chylomicrons

86. Haemoglobin has

- (1) Primary structure
(2) Secondary structure
(3) Tertiary structure
(4) Quaternary structure

87. Which is the most abundant biomolecule on earth?

- (1) Mineral salts
(2) Protein
(3) Lipids
(4) Carbohydrates

88. In which of the following an anticodon occurs

- (1) t-RNA (2) m-RNA
(3) r-RNA (4) DNA

89. The fastest enzyme is

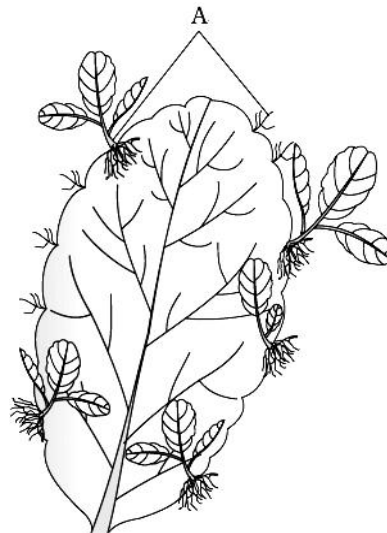
- (1) DNA gyrase
(2) Pepsin
(3) DNA polymerase
(4) Carbonic unhydrase

90. Which of the following is a phospholipid?

- (1) Sterol (2) Cholesterol
(3) Lecithin (4) Steroid

SPACE FOR ROUGH WORK

91. Adaptations which contributes in formation of end products of sexual reproduction in angiosperm :
(1) Inflorescences (2) Flowers
(3) Floral parts (4) All of these
92. Name the two parts (A) in a flower in which the two most important unit (B) of sexual reproduction develops
- | <u>A</u> | <u>B</u> |
|----------------------------|------------------------------------|
| (1) Sepal / Petal | Egg / pollen |
| (2) Stigma / Ovary | Egg / pollen |
| (3) Androecium / Gynoecium | Pollen / Egg |
| (4) Style / Filament | Pollen chamber/
Vascular supply |
93. The fundamental function of flowers is :
(1) Production of smell
(2) Production of progeny
(3) Symbol of feeling
(4) Make cultural value
94. Much before the actual flower is seen on a plant. Which hormone lead to the differentiation of floral primordium
(1) GA (2) IAA
(3) CK (4) Florigen
95. Sexual reproduction results in offspring that are :
(1) Identical to the parent due to law of segregation
(2) not identical to the parent due to law of segregation
(3) not identical to the parent due to mutation
(4) not identical to the parent due to law of dominance
96. Which one is multilayered in a typical anther :
(1) Epidermis (2) Endothecium
(3) Middle layer (4) Tapetum
97. Sporogenous tissue gets its nourishment from tapetum is :
(1) Megaspore mother cell / MMC
(2) Microspore mother cell / PMC
(3) Pollen grain / Microspore
(4) Egg cell
98. A mature dehiscent anther consisting of :
(1) Four pollen sac
(2) Two pollen sac
(3) Four pollen chamber
(4) Two megasporangia
99. Inside each microsporangia _____ of microspores / pollen grains are formed:
(1) Hundred
(2) Thousand
(3) Millions
(4) All of these
100. What is the function of germ pore :
(1) Faciliate the absorption of water on stigma
(2) Act as a passage of pollen tube
(3) Aperture through which the pollen tube emerges
(4) All of these
101. In crop breeding programmes, which type of banks are useful :
(1) Pollen bank (2) Seed bank
(3) Gene bank (4) All of these
102. Consider the given diagram -



Consider the correct term for given structure "A" and correctly matched with correct example of plant :

- (1) Adventitious bud → Ginger
(2) Axillary bud → Bryophyllum
(3) Axillary bud → Agave
(4) Adventitious bud → Bryophyllum

103. Which of the following type spores formation take place in fungi during asexual reproduction :
- Zoospores & conidia
 - Gemmules & conidia
 - Only conidia
 - Binary fission & conidia
104. Which of the following is not vegetatively propagate through the underground plant part :
- Potato
 - Sugarcane
 - Dahlia
 - Banana
105. If the number of chromosomes in a chlamydomonas cell is 8 then what is the number of chromosomes are present in zoospore produced during reproduction :
- 8
 - 4
 - 16
 - 12
106. 250 gm of a bacteria *Methylophilus methylotrophus* because of its high rate of biomass production and growth can produce
- 250 gram of protein per day
 - 200 gm of protein per day
 - 25 tonnes of protein per day
 - 25 tonnes of protein per week
107. The 'Roqueforti cheese' are ripened by a species of
- Aspergillus*
 - Yeast
 - Penicillium*
 - Bacteria
108. Penicillin was the first antibiotic to be discovered, it was a chance discovery while Alexander Flemming was working on
- Streptococcus*
 - Staphylococci*
 - Diplococcus*
 - Streptomyces*
109. Bioactive molecule cyclosporin A, which is used as an immuno suppressive agent, is produced by the fungus
- Trichoderma polysporum*
 - Monascus purpureus*
 - Penicillium*
 - Aspergillus niger*

110. Tissue plasminogen activator is
- A vitamin
 - A bioactive compound which is used as immunosuppressive agent
 - An enzyme
 - A chemical that activates tissue differentiation in tissue culture
111. Large holes in "Swiss cheese" are due to production of
- Large amount of CO₂ by *Propionibacterium sharmanii*
 - Large amount of CO₂ by *Monascus purpureus*
 - Small amount of Ethanol by *Penicillium*
 - Small amount of CO₂ by LAB
112. Identify the blank spaces A, B, C and D given in the following table and select the correct answer.

Type of Microbes	Scientific Name	Product	Medical Application
Fungus	A	Cyclosporin-A	B
C	<i>Monascus</i>	Statin	D

- A-*Trichoderma polysporum*, B-As an immunosuppressive agent in organ transplant agent
 - A-*Trichoderma polysporum*, B-As blood-cholesterol lowering agent, C-Protozoa, D-As an antibiotic
 - A-*Clostridium butylicum*, B-Used as a clot-buster, C-Yeast, D-As blood-cholesterol lowering agent
 - A-*Clostridium butylicum*, B-As blood-cholesterol lowering agent, C-Yeast, D-Used as a clot-buster
113. Wine and beer are produced
- Through distillation from barley malt
 - Through distillation from fruit juices
 - With or without distillation from Molasses
 - Without distillation from ripe fruit and barley malt

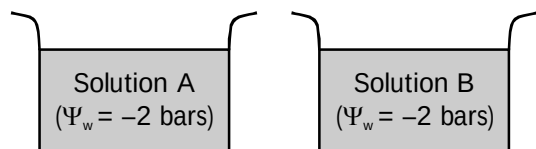
114. Probiotics are :-

- (1) Harmless amensal
- (2) Dietary supplements of useful microbes
- (3) Category of nondigestible functional food
- (4) They are non digestible food ingredient that stimulate the growth of bacteria in the digestive tract

115. You are monitoring the diffusion of coloured molecules across a membrane. Which of the following will result in the fastest rate of diffusion?

- (1) An internal concentration of 5% and an external concentration of 60%
- (2) An internal concentration of 35% and an external concentration of 40%
- (3) An internal concentration of 60% and an external concentration of 5%
- (4) Both (1) and (3) are correct

116.



Now, choose the correct option.

- (1) Kinetic energy (K.E.) of H_2O in A solution > K.E. of water in solution B
- (2) K.E. of water in solution B > K. E. of water in solution A
- (3) K.E. of water in solution A = K.E. of H_2O in solution B
- (4) Water potential has nothing to do with K.E. of water in a solution

117. The direction of the conduction of food through phloem is

- (1) from bottom to top
- (2) from top to bottom
- (3) Both of the above
- (4) phloem never conducts food

118. Minerals uptake can be depressed by depriving the root of O_2 which indicates ___(A)___ is required, such uptake involves ___(B)___ transport.

	A	B
(1)	CO_2	Passive
(2)	O_2	Facilitated transport
(3)	O_2	Active
(4)	H_2O	Antiport

119. An analysis of the plant sap indicates that

- (I) Some of the nitrogen travel in the organic compounds, much of it is carried as inorganic ions
- (II) Small amount of P and S are carried as organic compounds

- (1) I and II are correct
- (2) Both I and II are incorrect
- (3) Only I is correct
- (4) Only II is correct

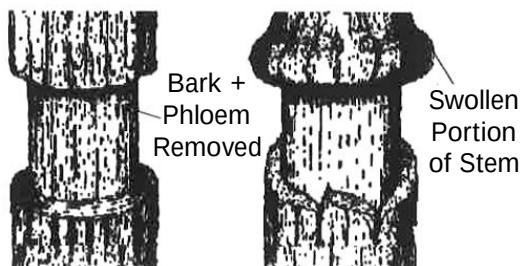
120. Scientists take advantage of in studying translocation by phloem

- (1) ants
- (2) aphids
- (3) bees
- (4) butterflies

121. Carbohydrates moves through the sieve as a result of the movement of water due to

- (1) cohesion
- (2) adhesion
- (3) diffusion
- (4) osmosis

122. The diagram refers the ringing or girdling experiment. Bark having phloem is removed. This experiment proves that phloem is the path for translocation of food. In this experiment swollen part of stem has been indicated. The possible cause of this swelling is



- (1) Accumulation of food materials just above the ring (As downward movement of food is inhibited)
(2) Accumulation of water and mineral just above the ring
(3) A repairing mechanism has taken place
(4) Injured parts undergo turgidity
123. When a plant is flooded it often increases its
(1) ethylene
(2) cytokines
(3) gibberellins
(4) All of these
124. Bulk flow can be achieved through a hydrostatic pressure gradient. Negative hydrostatic pressure gradient and positive hydrostatic pressure gradient are seen in :-
(1) Suction through a straw and a garden hose respectively
(2) A garden hose and suction through a straw respectively
(3) Cobalt chloride paper and polyethene respectively
(4) Blotting paper and cobalt chloride paper respectively
125. The force of tension-cohesion exceeds root pressure on
(1) rainy day
(2) fog morning
(3) sunny day
(4) full moon night

126.

	Column-I		Column-II
(I)	IAA	(A)	Terpenes
(II)	GA	(B)	Indole compounds
(III)	ABA	(C)	Adenine derivatives
(IV)	C_2H_4	(D)	Gases
(V)	Kinetin	(E)	Carotenoid derivatives

The correct match is :-

- (1) I-B, II-A, III-E, IV-D, V-C
(2) I-A, II-B, III-C, IV-D, V-E
(3) I-E, II-D, III-A, IV-B, V-C
(4) None of these
127. Statement I : Confirmation of the released of volatile substances from ripened oranges that hastened the ripening of stored unripe bananas.
Statement II : Callus formation from the internodal segments of tobacco stem with the Auxin and extracts of vascular tissue/ yeast/coconut milk/DNA. Later, cytokinesis promoting substance was identified, crystallized and named as Kinetin.
Statement III : Reporting of appearance of symptoms of bakane/follicle seeding disease caused by fungus, *Gibberella fujikuroi*, in uninfected seedling when they were treated with sterile filtrates of the fungus. the active substances were later identified as GA.
Choose the correct option.

	I	II	III
(1)	Miller and Skoog	Cousins	Kurosawa
(2)	Kurosawa	Cousins	Miller and Skoog
(3)	Cousins	Kurosawa	Miller and Skoog
(4)	Cousins	Miller and Skoog	Kurosawa

128. Auxin :-

- (1) is not transported, because auxin is used where it is made
- (2) is transported by diffusion with no preferred direction
- (3) is transported from base to tip
- (4) is produced by the growing apices of the stem and roots, from where they migrate to the regions of their action

129. (I) "Y" hormone induced flowering in mango and also promotes rapid internode/petiole elongation in deep water rice plants and hence helping leaves or upper part of shoot above water

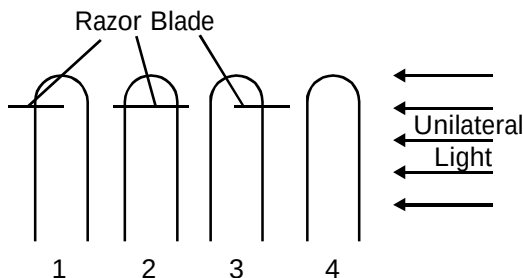
(II) "X" hormone promotes root growth and root hair formation

(III) "Z" hormone inhibits the seed germination, increases the tolerance of plant to various stresses, play important role in seed development, maturation and dormancy.

Identify the correct names of the hormones.

- (1) Y = ABA; X = Auxin; Z = GA
- (2) Y = GA; X = Auxin; Z = C_2H_4
- (3) Y = Auxin; X = C_2H_4 ; Z = GAA
- (4) Y = C_2H_4 ; X = C_2H_4 ; Z = ABA

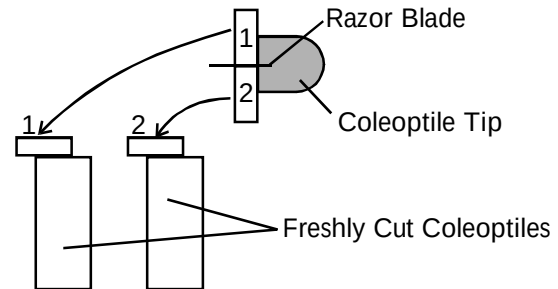
130. The following diagram shows four coleoptiles set up at the start of an experiment.



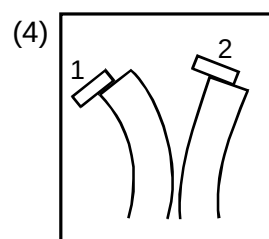
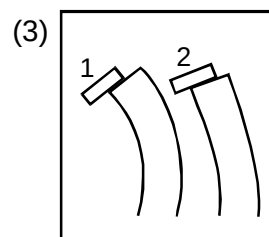
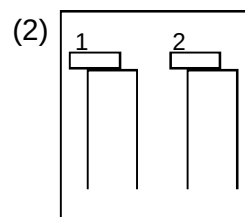
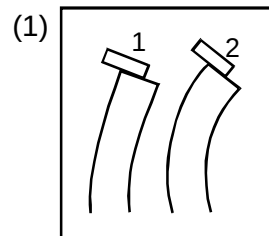
Which two coleoptiles bend towards the light sources?

- (1) 1 and 2
- (2) 1 and 4
- (3) 2 and 3
- (4) 3 and 4

131. Agar block 1 and 2 were kept in the positions shown in the diagram below for several hours and then transferred on to two freshly cut coleoptiles.

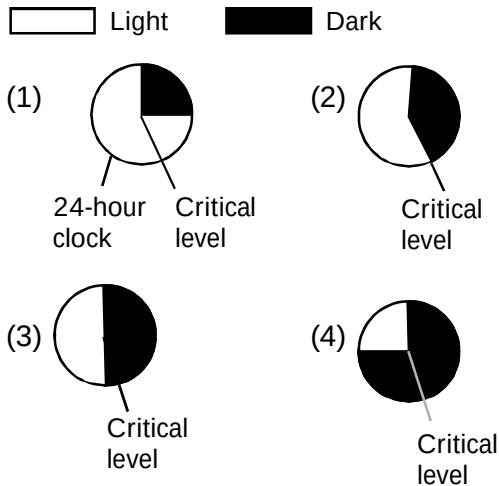


Which of the following would result after two days of growth ?



132. Maryland mammoth tobacco is a short day plant. Its critical duration of darkness is 10 hours.

Under which of the following conditions will it not flower?



133. Clinostat is the apparatus used to :-

- (1) Measure the rate of growth in plant
- (2) Measure the quantity of auxin in plant
- (3) Measure the effect of light on plant
- (4) Eliminate the effect of gravity or geotropism on plant

134. Read the following statements and find out the incorrect statement :-

- a. In plants, growth and differentiation both are open, i.e it can be indeterminated or determinate.
- b. The final structure at maturity of a cell or tissue is also determined by the location of the cell within.
- c. Cells positioned away from root apical meristems mature and epidermis, while those pushed to the periphery differentiate as root cap cells.
- d. Environmental signals such as light, temperature and gravity also effect certain phases or stages of growth.
- e. Intrinsic factors includes both intercellular (genetic) or intracellular factors (chemicals such as plant growth regulators) while the extrinsic factors includes light, temperature, water, oxygen, nutrition, etc.

- (1) b and d
- (2) c and a
- (3) a and e
- (4) c and e

135. Fill in the blanks :-

1. Spraying sugarcane crop with a increases in the length of the stem, thus increasing the yield by as much as b tonnes per acre.
2. c does not occur naturally in plants.
3. Search for natural substances with cytokinin like activities led to the isolation of d from corn kernels and coconut milk.

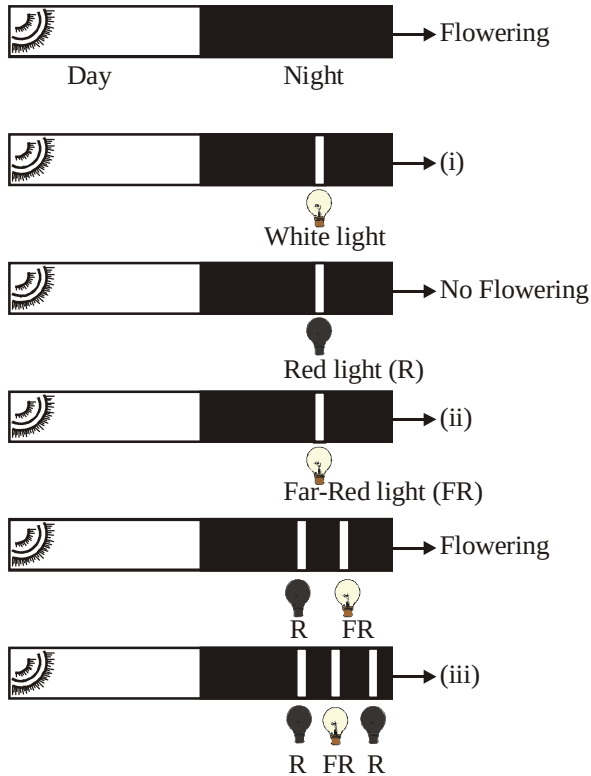
- (1) a-auxins, b-10, c-NA A, d-zeatin
- (2) a-gibberellins, b-20, c-zeatin, d-kinetin
- (3) a-gibberellins, b-10, c-zeatin, d-kinetin
- (4) a-gibberellins, b-20, c-kinetin, d-zeatin

136. Fill in the blanks :-

1. Wheat, barley and rye have two kinds of varieties; winter and spring varieties. The 'spring' variety are normally planted in a and come to flower and produce grain before the end of growing season.
2. 'Winter' varieties, however, if planted in b would normally fail to flower or produced mature grain within a span of flowering season.
3. Hence 'winter' varieties are planted in c. They germinate, and over d come out as small seeding resume growth in the e and are harvested usually around mid-summer.

- (1) a-spring, b-winter, c-spring, d-winter, e-spring
- (2) a-winter, b-spring, c-winter, d-spring, e-winter
- (3) a-spring, b-spring, c-autumn, d-winter, e-spring
- (4) a-spring, b-winter, c-autumn, d-spring, e-winter

137. Given figure shows the effect of interruption on scotoperiod (dark period) in a short day plant by light of different types.



Select the correct option for (i), (ii) and (iii)

	(i)	(ii)	(iii)
(1)	Flowering	Flowering	No flowering
(2)	No flowering	No flowering	Flowering
(3)	No flowering	Flowering	No flowering
(4)	Flowering	No flowering	No flowering

138. Sexual reproduction is character of :

- (1) Flowering plant only
- (2) Embryophyta only
- (3) All cryptogamous only
- (4) Almost all plants

139. Siphonogamous sexual reproduction is the character of

- (1) Tracheophyta
- (2) Spermatophyta
- (3) Embryophyta
- (4) Thallophyta

140. Sexual reproduction is absent in :

- (1) Lower plants, Lichen
- (2) Rhodophyceae, Blue green algae
- (3) Monera, Rhodophyceae
- (4) Monera, Deuteromycetes

141. According to phylogeny & evolutionary trends sexual reproduction in flowering plant is the advanced form of :

- (1) Isogamy
- (2) Anisogamy
- (3) Oogamy
- (4) Parthenogamy

142. As compared to asexual reproduction, sexual reproduction is :

- (1) an elaborate, complex and rapid process
- (2) an elaborate, complex and slow process
- (3) an elaborate, narrow and rapid process
- (4) a simple, complex and rapid process

143. Odd one is :

- (1) Stamen - Microsporophyll
- (2) Microsporangia - Pollen sac
- (3) Microspore - Pollen
- (4) Endothecium - Nutritive tissue

144. How tapetal cells could become bi-nucleate :

- (1) By free nuclear mitosis
- (2) By amitosis
- (3) By endomitosis
- (4) By meiosis

145. Type of meiosis by which sporogenous tissue form microspore tetrad :

- (1) Zygotic
- (2) Sporic
- (3) Gametic
- (4) All

146. If total 25 cells are present in the sporogenous tissue of microsporangia. How many microspore tetrad will come in existence :

- (1) 100
- (2) 25
- (3) 125
- (4) 250

147. In Hibiscus, yellowish colour of pollen grains :

- (1) due to xanthophylls
- (2) due to pollen kit
- (3) due to phycobillins
- (4) due to fucoxanthin

148. When pollen grain is mature it contains :
 (1) One cell (2) Two cells
 (3) Three cells (4) Four cells
149. The male gametes of angiosperms are :
 (1) Uniflagellate (2) Bi-flagellate
 (3) Non motile (4) None of these
150. Which of the following mode of reproduction is better for organisms :
 (1) Asexual
 (2) Sexual
 (3) Vegetative propagation
 (4) Apomixis
151. Which of the following are microscopic motile spores
 (1) Zoospores (2) Conidia
 (3) Bud (4) Gemmules
152. Plant which is known as "Terror of Bengal" is generally reproduce by :
 (1) Underground stem - Rhizome
 (2) Formation of zoospores
 (3) Sub-aerial stem - offset
 (4) Aerial stem - Phylloclade
153. Biogas is mixture of mainly
 (1) CO_2 , CO , N_2
 (2) CH_4 , H_2 , C_2H_4
 (3) CH_4 , CO_2 , H_2S , H_2
 (4) CH_4 , SO_2 , H_2
154. Which of the following is correct for Mycorrhiza?
 (1) Absorb nitrogen and water only
 (2) Restricted to flowering plants only
 (3) Symbiotic association of fungi with roots of plants which fix nitrogen and hence regarded as biofertilizer
 (4) Many members of genus *Glomus* form Mycorrhiza
155. Mark the mismatched pair.
 (1) *Azolla* - important biofertilizer in paddy field
 (2) Nucleopolyhedrovirus - IPM
 (3) *Azospirillum* - Free living N_2 fixer in soil
 (4) Rotenones - obtained from roots of *Derris elliptica*
156. Using spores of *Bacillus thuringiensis* for pest control is
 (1) IPM
 (2) Biological control
 (3) Concept of sustainable agriculture
 (4) All of these
157. An example of microbial biocontrol agent that can be introduced in order to control butterfly caterpillars is
 (1) Lady bird beetle
 (2) *Bacillus thuringiensis*
 (3) *Streptococcus*
 (4) Dragonflies
158. Match the following.
 (A) Lady bird beetle (i) Mosquito
 (B) Dragon flies (ii) Insects or other arthropods
 (C) Trichoderma (iii) Aphids
 (D) Baculovirus (iv) Plant pathogens
 (1) A(iii), B(i), C(iv), D(ii)
 (2) A(iii), B(iv), C(i), D(ii)
 (3) A(i), B(iii), C(ii), D(iv)
 (4) A(iv), B(i), C(iii), D(ii)
159. Find the incorrect statement w.r.t antibiotics:
 (1) An antibiotic is a substance produced by an organism which in low concentration inhibit the growth and metabolic activity of pathogenic organisms
 (2) Produced by many microbes
 (3) Waksman discovered sulphadiazine and coined the term antibiotics
 (4) Antibiotic have greatly improved our capacity of treating deadly diseases

160. Consider the following statements about *Bt*.
- The bacteria *Bacillus thuringiensis* (*Bt*) are used to control butterfly caterpillars.
 - Fresh spores of *Bt* are mixed with water and sprayed on seeds such as brassicas and fruit trees.
 - Insect larvae, after eating these are killed by the toxin released in their bladder.
 - Bt* toxin genes have been introduced into plants to provide resistance to pests.
- Which of the statements given above are correct?
- I, II and III
 - I, II and IV
 - II, III and IV
 - I, II, III and IV
161. Which is not a true statement regarding macronutrients ?
- Form plant structure
 - Become toxic in excess
 - Role in metabolism
 - Develop osmotic potential
162. Releasing of NH_3 from organic nitrogen by the activity of microbes is called :-
- Nitrogen fixation
 - Nitrification
 - Proteolysis
 - Ammonification
163. Which of the following pairs is odd ?
- Fe, Cu
 - Zn, Mn
 - Ca, Mg
 - B, Cl
164. In biological nitrogen fixation, how many ATP are required for each NH_3 produced ?
- 6 ATP
 - 8 ATP
 - 12 ATP
 - 16 ATP
165. Full form of BOD.
- Biological Oxygen Demand
 - Bio Oxygen Demand
 - Biochemical Oxygen Demand
 - Biochemical Oxygen Degradable
166. Which of the following is the contribution of bacteria during symbiotic nitrogen fixation ?
- Auxin hormone
 - Haem part of leghaemoglobin
 - ATP
 - Globin part of leghaemoglobin
167. Which of the following elements is essential for photosynthesis as well as helps to maintain the structure of protein factory of the cell ?
- Mg
 - N
 - Fe
 - Ca
168. Which of the following are not trace elements ?
- Boron and Manganese
 - Copper and Zinc
 - Calcium and Sulphur
 - Chlorine and Molybdenum
169. Which of the following statements is true ?
- Every mineral element, that is present in a cell is essential for cell
 - Boron deficiency leads to stout axis
 - Nitrogen as a nutrient element is highly immobile in plants
 - It is very easy to establish the essentiality of trace elements, because they are required only in minute quantity
170. A - Uptake and utilisation of Ca^{2+}
B - Pollen tube formation
C - Activator of carboxylase
D - Nitrogen metabolism
E - Oil synthesis
F - Carbohydrate translocation
- How many of the above functions are performed by boron ?
- Two
 - Three
 - Four
 - Five

171. Which of the following statements regarding baculoviruses as bio-control agents is/are correct ?

- I. Baculoviruses are pathogens that attack insects and other arthropods.
- II. Most of these biocontrol agents belong to the genus *Nucleopolyhedro* virus.
- III. They do harm plants, mammals, birds, fish and other non-target insects.
- IV. Baculoviruses are helpful in Integrated pest Management (IPM) programme, in which beneficial insects are conserved.

Choose the correct option.

- (1) I, II and III
- (2) I, II and IV
- (3) II, III and IV
- (4) All of these

172. Which one of the following is not a biopesticide?

- (1) *Bacillus thuringiensis*
- (2) Nucleopolyhedro virus
- (3) Trichoderma
- (4) Agrobacterium

173. Biogas is the mixture of gases produced by the microbial activity. The type of the gas produced depends upon :-

- (1) Type of microbes
- (2) Type of organic substrate/waste
- (3) Size of digester
- (4) Both (1) and (2)

174. Which of the following bioactive molecule has been commercialised as blood cholesterol lowering agent ?

- (1) Cyclosporin-A
- (2) Streptokinase
- (3) Statin
- (4) All of the above

175. Which of the following alcoholic drinks are produced by distillation of the fermented broth ?

- (1) Wine
- (2) Whisky
- (3) Rum
- (4) (2) & (3) Both

176. Biogas is a mixture of gases produced by the microbial activity in which predominant gas is :-

- (1) CO₂
- (2) CO
- (3) Methane
- (4) H₂

177. Identify the correct match from the Column-I and Column-II :-

	Column-I		Column-II
a	<i>Aspergillus niger</i>	i	Butyric acid
b	<i>Acetobacter aceti</i>	ii	Citric acid
c	<i>Clostridium butylicum</i>	iii	Lactic acid
d	<i>Lactobacillus</i>	iv	Acetic acid

(1) a-ii, b-iv, c-i, d-iii

(2) a-iii, b-iv, c-i, d-ii

(3) a-ii, b-i, c-iv, d-iii

(4) a-iv, b-ii, c-i, d-iii

178. The dough, which is used for making bread is fermented using :-

- (1) *saccharomyces cerevisiae*
- (2) *propionibacterium sharmanii*
- (3) *acetobacter aceti*
- (4) *streptococcus*

179. Enzyme which is produced by the bacterium *streptococcus* and modified by genetic engineering is used as a "clot buster" for removing clots from the blood vessels of patients who have undergone myocardial infarction leading to heart attack is :-

- (1) Pectinases
- (2) Streptokinase
- (3) Polygalacturonase
- (4) Adenosine deaminase

180. Which of the following organism serve as an important biofertiliser in paddy fields ?

- (1) Fungi
- (2) Fern
- (3) Beetle
- (4) None of these

Space For Rough Work

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